

AEO Daily Session Health — Process Guide



Signal AEO Dashboard
(every 14 days)

Platforms: ChatGPT · Gemini · Perplexity

Ranking cycle: bi-weekly

1. WHAT IS DAILY SESSION HEALTH MONITORING?

Core concept

AI answer engines (ChatGPT, Gemini, Perplexity) learn which businesses to recommend from **real user query interactions**. Every session your automation runs is one signal telling the engine "this business is relevant for this keyword in this city." Daily health monitoring tracks whether enough of those signals are being sent — consistently, across all keywords and all platforms — before the bi-weekly ranking run captures the result.

Without monitoring, you would only find out that sessions failed or keywords were missed **after** the ranking run drops. With daily health monitoring, you catch and fix problems with enough time to recover before the next measurement.

2. THE 7 HEALTH CRITERIA — WHAT GETS MEASURED EVERY DAY



3 — Rolling 7-Day Average

The 7-day smoothed view of sessions/day and success rate. More reliable than a single day's snapshot.

- **Avg sessions:** $\geq$ 80% of target/day
- **Avg success:** $\geq$ 90%
- **Gap days:** Zero-session days tracked (last 7)

Why it matters: One good day after 6 bad days still shows a weak 7-day average. Engines weigh recency AND consistency — the 7-day average is the closest proxy to what the engine "remembers."



4 — Keyword Coverage

Each keyword must be hit regularly. Engines track keyword coverage over time since each keyword last hit.

- **OK:** Hit today or yesterday
- **Warning:** 3+ days since last hit
- **Critical:** 5+ days since last hit

Why it matters: A keyword with no hits is cold. The engine may stop associating it with your site. This is often the first visible cause of a drop in rankings.



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Prediction for Next Ranking Run

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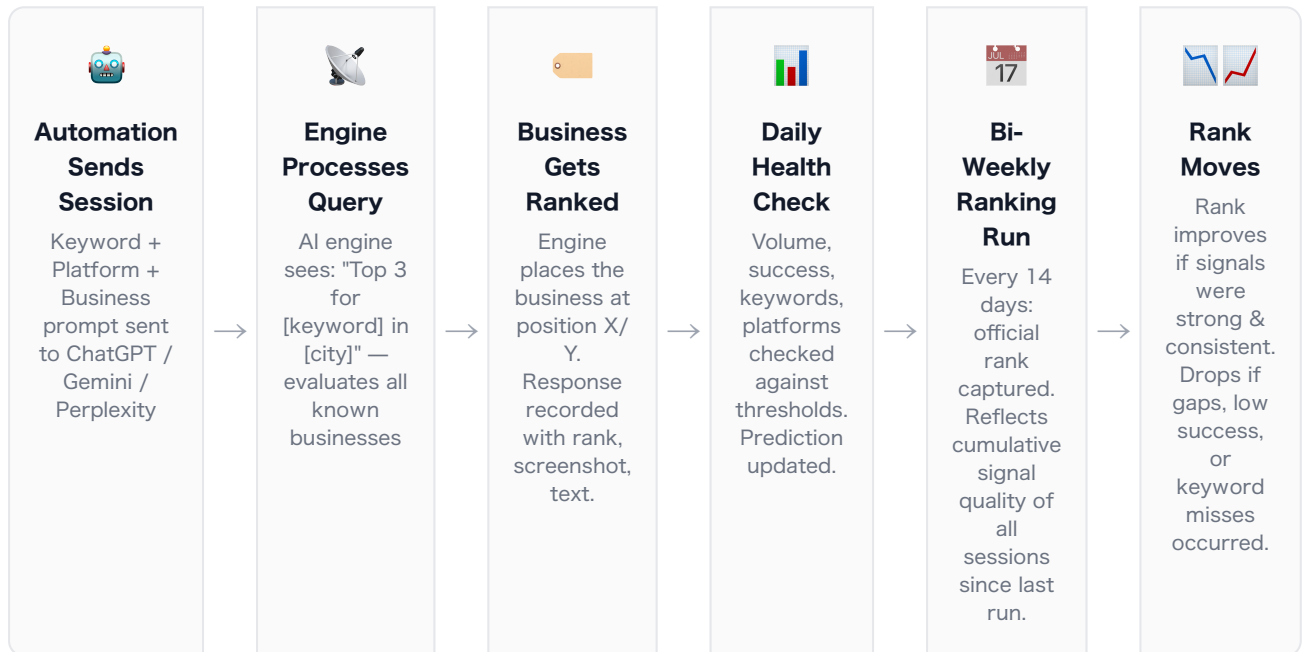
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2. HOW EACH CRITERION DRIVES AEO RANKING IMPROVEMENT

HEALTH SIGNAL	WHAT BREAKS WHEN IT FAILS	WHAT YOU DO TO FIX IT	IMPACT ON RANKING
Session Volume below target	Not enough daily reinforcement. Engine confidence in the business drops over the 14-day window.	Check automation queue depth. If queue is backed up: scale workers or increase concurrency. If empty: check campaign scheduler for missed triggers.	Medium Slow decline over 2–3 weeks
Success rate < 70%	Majority of sessions produce no signal. Proxy failure, rate-limiting, or prompt errors cause "silent" sessions that waste budget.	1. Check proxy pool health (rotation, city targeting). 2. Review error messages in CSV error column. 3. Test prompt variant on failing platform. 4. Rotate proxy city if 3+ consecutive failures from same IP.	High Rapid drop if sustained 3+ days
Keyword missed 3+ days	Engine stops receiving "this business → this keyword" association. Ranking for that specific query weakens.	Identify why keyword is being skipped: check campaign keyword list for typos, check if keyword has a variant map issue, manually trigger a test session for that keyword.	Medium Keyword-level drop within 1–2 ranking cycles
Keyword missed 5+ days	Keyword is effectively cold. The engine treats the business as not relevant for that query.	Immediate manual session push for that keyword across all 3 platforms. After recovery, add keyword to daily priority list for 7 days.	High Keyword-level "NOT FOUND" at next ranking run
Platform silent 3–4 days	That platform's index starts to forget the business. Other platforms may still be fine.	Check if the platform-specific automation script is erroring (check logs). Test a single manual session on the silent platform to confirm it's working.	Medium Platform-specific drop, not portfolio-wide
Platform silent 5+ days	Platform index has lost the business's signal momentum. Recovery takes 5–7 days of heavy sessions.	Double session target on the silent platform for 5 days. Do not just resume normal volume — you need catch-up volume to rebuild the signal.	High Likely "SUDDEN DROP" or "NOT FOUND" on platform
Gap day (0 sessions)	Breaks consistency. Engine deprioritizes businesses that "disappear" even for one	Investigate why: server down, scheduler missed, RabbitMQ queue empty? Next day: push	Medium Single gap tolerated; 2+

HEALTH SIGNAL	WHAT BREAKS WHEN IT FAILS	WHAT YOU DO TO FIX IT	IMPACT ON RANKING
	day during active ranking cycles.	1.5× normal volume to compensate for the missed day.	gaps = at-risk territory
Prediction: AT_RISK	Multiple criteria failing simultaneously. Compound effect — each failure amplifies the others.	Follow the 4-Week Ideal Flow (Section 4). Week 1: diagnose root cause. Don't just add volume — fix the underlying issue first, then push volume.	Critical Expect ranking drop at next bi-weekly run

3. HOW THE SIGNAL CHAIN WORKS — SESSION → RANKING



The 14-day window is critical

Everything that happens between two ranking runs is what determines the next rank. A problem on Day 3 of the cycle that goes unfixed means 11 more days of weakened signal. That's why daily monitoring exists — to catch problems on Day 3, not Day 14.

4. THE 4-WEEK IDEAL FLOW — FROM AT_RISK TO ON_TRACK

When a business shows **AT_RISK**, follow this 4-week recovery process:

WEEK 1	WEEKS 2-3	WEEK 3-4	WEEK 4
🔍 Diagnose Root Cause • Open Health Monitor → find the business • Check which criteria are failing (volume? success? keyword? platform?) • Read the Why field — it states the specific issue • Check the CSV error column for that business/date • Identify if it's a proxy issue, scheduler issue, or campaign config issue	🔧 Fix + Push Volume • Apply the fix identified in Week 1 • Run a test session on each failing keyword/platform • Once success rate $\geq 90\%$, push 1.5x normal daily volume • Monitor keyword coverage daily — all keywords should hit every 1-2 days • Ensure all 3 platforms receive sessions every day • Check daily that prediction moves from	📊 Stabilize & Monitor • Return to normal daily volume once 7-day average is $\geq 90\%$ • Confirm: 0 gap days in last 7 • Confirm: all keywords hit within last 2 days • Confirm: all 3 platforms active last 3 days • Prediction should now show ON_TRACK  • Document root cause for LLM training data	📝 Re-measure Results • Next bi-weekly ranking run captures the result • Compare position to pre-AT_RISK baseline • If improved: record what worked → use as LLM training example • If still down: review health data again — likely a platform-specific issue or a keyword that was cold longer than expected • Adjust campaign target if needed (increase target/day)

5. PLATFORM-SPECIFIC STRATEGY

PLATFORM	SIGNAL CHARACTERISTICS	WHAT HURTS RANKINGS HERE	RECOVERY ACTIONS
ChatGPT push more volume yet fix the root cause first	Tends to weight business description quality and Google Maps listing consistency. More forgiving of short gaps.	Low success rate (rate-limiting common). Keyword prompts that don't match GPT's style. Proxy IPs flagged as bots.	Rotate prompt variants. Use residential proxies. If success $< 70\%$ for 2 days, switch prompt to more natural language format.
Gemini	Strongly weights recency and local authority signals. Most sensitive to gap days — loses memory faster than other platforms.	Any gap day drops Gemini faster than others. Low keyword diversity (only one keyword)	Zero tolerance for Gemini gaps. If 2 consecutive Gemini gap days: treat as critical and push 2x volume immediately. Use all keyword variants.

PLATFORM	SIGNAL CHARACTERISTICS	WHAT HURTS RANKINGS HERE	RECOVERY ACTIONS
		variant used repeatedly).	
Perplexity	Citation-based — weights how often the business appears in search results that Perplexity indexes. Most keyword-sensitive platform.	Keywords not matching the exact phrasing Perplexity indexes. Business not appearing in cited web sources.	Expand keyword variants — Perplexity responds to more natural long-tail phrases. If "NOT FOUND" appears: check that the Google Maps listing is indexable and current.

6. KEYWORD COVERAGE — HOW TO MAINTAIN FULL COVERAGE

⚠ Most ranking drops start with keyword gaps, not volume drops

A business can have 100% volume and 95% success rate, but if one keyword hasn't been hit for 5 days, that keyword's ranking will drop at the next run — even if every other keyword is perfect.

Keyword rotation rules

- Each canonical keyword has a set of **variant phrases** used in the prompt (e.g., "best mobile detailing" → "top mobile detailing," "mobile car detailing near me," etc.)
- The system tracks the **canonical keyword** — any successful session with any variant counts as a hit
- A keyword is marked **OK** if it was hit in the last 1–2 days
- At **3 days** without a hit → show ⚠ warning — manual check needed
- At **5 days** without a hit → 🚨 critical — push a targeted session that day

Example keyword health states:

best mobile detailing — Today ✓	car wash at home — 1 d ago ✓
mobile detailing service — 3d ago ⚠	before and after detailing — 5d ago 🚨

Action rule: When you see a 🚨 critical keyword, do not wait for the automated queue. Manually trigger a session for that specific keyword across all 3 platforms before end of business day.

7. HOW DAILY HEALTH DATA TRAINS THE AEO LLM

The daily health monitoring data is the **ground truth** used to train and fine-tune the AEO LLM. Each business's session history becomes a labeled training example that teaches the model to diagnose and prescribe actions.

1

Input Features (X)

Raw session signals captured every day from the CSV:

- Sessions today vs. target
- Success rate (today + 7-day avg)
- Keywords hit today
- Days since each keyword last hit
- Platform session counts (3-day window)
- Gap days in last 7
- Days active / campaign phase
- Current ranking label (STABLE, SUDDEN_DROP...)

2

Labels (Y)

The diagnostic output the model learns to produce:

- **Prediction:** ON_TRACK / AT_RISK / STABLE / TOO_EARLY
- **Why:** Plain-English diagnosis of the root cause
- **Action:** Specific next step for the account manager

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Why: "7-day success rate 64%  
- proxy failures on Gemini  
for keyword 'mobile detailing'"  
  
Action: "Rotate Gemini proxy pool,  
push 2x sessions on this keyword  
for next 5 days"
```

3

LLM Output

After training, the model can:

- Predict health state from raw daily data before a human reviews it
- Generate the **Why** explanation automatically
- Prescribe a ranked list of **Actions** in priority order
- Answer natural-language questions: "Why did X drop on Gemini this week?"
- Correlate sessions to ranking outcomes across the 14-day cycle
- Flag new businesses whose early session

The feedback loop that makes the LLM smarter over time

STEP	WHAT HAPPENS	HOW IT IMPROVES THE MODEL
1 — Capture	Every session is saved to the consolidated daily CSV with timestamp, business, keyword, platform, status, rank, and error fields.	Creates a timestamped audit trail that can be replayed to reconstruct any day's health state for any business.
2 — Score	The <code>csv-data.ts</code> library computes health metrics (volume %, success rate, keyword coverage, platform windows) and assigns the prediction label.	Generates the structured feature vector (X) used as model input. Same logic the model learns to replicate.
3 — Label	The <code>why</code> and <code>action</code> strings are computed by rule-based logic today. Over time, human reviewers annotate edge cases with better explanations.	Human-annotated examples are the highest-quality training labels. Even 50 well-annotated edge cases dramatically improve model diagnostic quality.
4 — Verify	14 days later, the ranking run captures actual outcomes. The system can compare: "we predicted AT_RISK on Day 3 — did the rank actually drop?"	Creates verified outcome labels. A prediction that was correct (AT_RISK → rank dropped) is a high-confidence training example. Incorrect predictions reveal model weaknesses.
5 — Retrain	Monthly or bi-monthly, collect verified examples → fine-tune the model on new patterns (new platforms, new campaign types, new proxy failure modes).	Model continuously improves — each ranking cycle adds ~N_businesses × 14 new training examples automatically.

8. DAILY WORKFLOW FOR ACCOUNT MANAGERS

✅ What to do every morning (takes 5–10 minutes)

#	ACTION	WHERE TO LOOK	IF YOU SEE A PROBLEM
1	Open Health Monitor. Check summary tiles: how many AT_RISK today?	Dashboard → Session Health Monitor → summary screen	If AT_RISK count increased vs. yesterday → start there first
2	Filter by AT_RISK. For each business: read the "Why" and "Action" fields.	Left sidebar → click AT_RISK filter → click each business	The Action field tells you exactly what to do. Follow it.
3	Check STABLE businesses. Look for platform windows with 3+ days silent.	Health Monitor → filter STABLE → expand Platform Rotation section	A platform at "⚠️ No sessions 3d" on a STABLE business = it's

#	ACTION	WHERE TO LOOK	IF YOU SEE A PROBLEM
			about to become AT_RISK. Treat it now.
4	Scan keyword coverage table for any 🚨 Critical keywords (5+ days).	Click any business → scroll to Keyword Coverage section	Manually trigger a session for that keyword today. Do not wait for the queue.
5	Check overall success rate in the daily HTML report.	Navigate to /api/csv/daily-report?date=YYYY-MM-DD	Any business <80% success → investigate proxy/prompt issue before pushing more volume
6	For businesses approaching their next ranking run (within 3 days): confirm ON_TRACK prediction.	Health Monitor → check "Next Ranking Run" date in each business header	If AT_RISK within 3 days of ranking run → escalate, push max volume, fix immediately

9. QUICK REFERENCE — PREDICTION STATES & RESPONSES

PREDICTION	WHAT IT MEANS	URGENCY	STANDARD RESPONSE
 ON_TRACK	All health criteria green. Business is building signal momentum toward the next ranking run.	None	No action needed. Maintain current volume and keyword rotation. Review daily to catch any regression.
 STABLE	Borderline. Mostly healthy but at least one criterion is borderline (e.g., 7-day avg slightly below target, or one platform at 3-day warning).	Low — Monitor	No immediate action but watch daily. If borderline issue not resolved within 2 days → treat as AT_RISK.
 AT_RISK	One or more criteria in critical failure. Ranking drop is likely at the next bi-weekly run without intervention today.	High — Act Today	Follow the 4-Week Ideal Flow. Read the "Why" and "Action" fields. Fix root cause first, then increase volume.
 TOO_EARLY	Less than 7 days of data. Prediction model cannot produce a reliable assessment yet.	Info Only	Confirm sessions are running and succeeding. If the latest session date is 3+ days old → restart sessions immediately (the business has stalled, not just started).